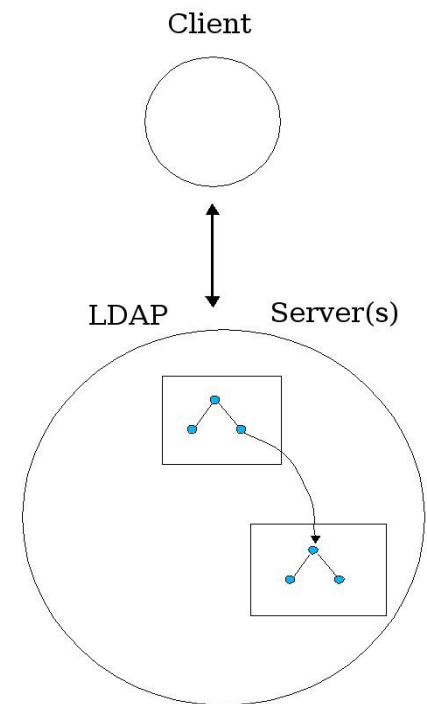


LIGHT WEIGHT DIRECTORY ACCESS PROTOCOL (LDAP)

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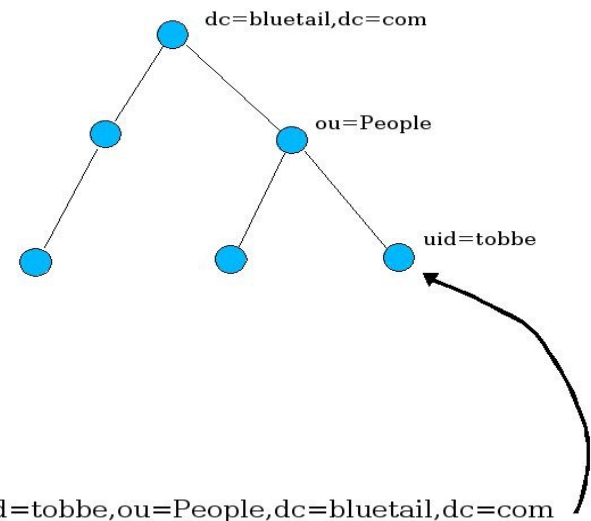
What is LDAP ?

- ▶ LDAP provides directory access, a centralized database of information about people, groups and other entities.
- ▶ Defined as a set of protocol operations against servers.
- ▶ Assumes one (or more) servers which jointly provide access to the DIT (Directory Information Tree)
 - ▶ The directory structure is a specialized database which is optimized for browsing, searching, locating and reading information.
- ▶ It is an internet protocol runs over TCP/IP.
- ▶ It is an open protocol, applications need not to worry about the type of server hosting the directory.
- ▶ Protocol described in ASN.1



The Directory Information Tree (DIT)

- ▶ The DIT is made up of entries.
- ▶ Entries have names consisting of one (or more) attribute values.
- ▶ The concatenation of the entry names form a path, the Distinguished Name (DN), which uniquely identifies an entry.



Distinguished Name (DN)

- ▶ In LDAP (Lightweight Directory Access Protocol), the directory tree is addressed using a hierarchical structure known as the Distinguished Name (DN). The DN represents the unique path or location of an entry within the LDAP directory tree.
- ▶ The DN is composed of multiple Relative Distinguished Names (RDNs) separated by commas.
 - ▶ Each RDN consists of an attribute and its corresponding value.
 - ▶ The order of RDNs in the DN represents the path from the entry to the root of the directory tree.
- ▶ For example, consider the following DN:
 - ▶ `cn=John Doe, ou=Users, dc=example, dc=com`
- ▶ In this DN:
 - ▶ `cn=John Doe` represents the Common Name (cn) attribute with the value "John Doe".
 - ▶ `ou=Users` represents the Organizational Unit (ou) attribute with the value "Users".
 - ▶ `dc=example` represents the Domain Component (dc) attribute with the value "example".
 - ▶ `dc=com` represents the Domain Component (dc) attribute with the value "com".
- ▶ Together, these RDNs form the complete DN that uniquely identifies the entry "John Doe" within the LDAP directory tree.

LDAP Functions/Operations

- ▶ Authentication
 - ▶ BIND/UNBIND
 - ▶ ABANDON
- ▶ Query
 - ▶ Search
 - ▶ Compare entry
- ▶ Update
 - ▶ Add an entry
 - ▶ Delete an entry (Only Leaf nodes, no aliases)
 - ▶ Modify an entry, Modify DN/RDN

Client and Server Interaction

- ▶ Client establishes session with server (BIND)
 - ▶ Hostname/IP and port number
 - ▶ Security
 - ▶ User-id/password based authentication
 - ▶ Anonymous connection - default access rights
 - ▶ Encryption/Kerberos also supported
- ▶ Client performs operations
 - ▶ Read/Update/Search
 - ▶ `SELECT X,Y,Z FROM PART_OF_DIRECTORY`
- ▶ Client ends the session (UNBIND)
- ▶ Client can ABANDON the session

BIND/UNBIND/ABANDON

- ▶ Request includes LDAP version, the name the client wants to bind as, authentication type
 - ▶ Simple (clear text passwords, anonymous)
 - ▶ Kerberos v4 to the LDAP server (krbv42LDAP)
 - ▶ Kerberos v4 to the DSA server (krbv42DSA)
- ▶ Server responds with a status indication
- ▶ UNBIND: Terminates a protocol session
 - ▶ UnbindRequest ::= [APPLICATION 2] NULL
- ▶ ABANDON:
 - ▶ MessageID to abandon

Search/Compare

- ▶ Request includes
 - ▶ baseObject: an LDAPDN
 - ▶ Scope: how many levels to be searched
 - ▶ derefAliases: handling of aliases
 - ▶ sizeLimit: max number of entries returned
 - ▶ timeLimit: max time allowed for search
 - ▶ attrsOnly: return attribute types OR values also
 - ▶ Filter: cond. to be fulfilled when searching
 - ▶ Attributes: List of entry's attributes to be returned
- ▶ Read and List implemented as searches
- ▶ Compare: similar to search but returns T/F

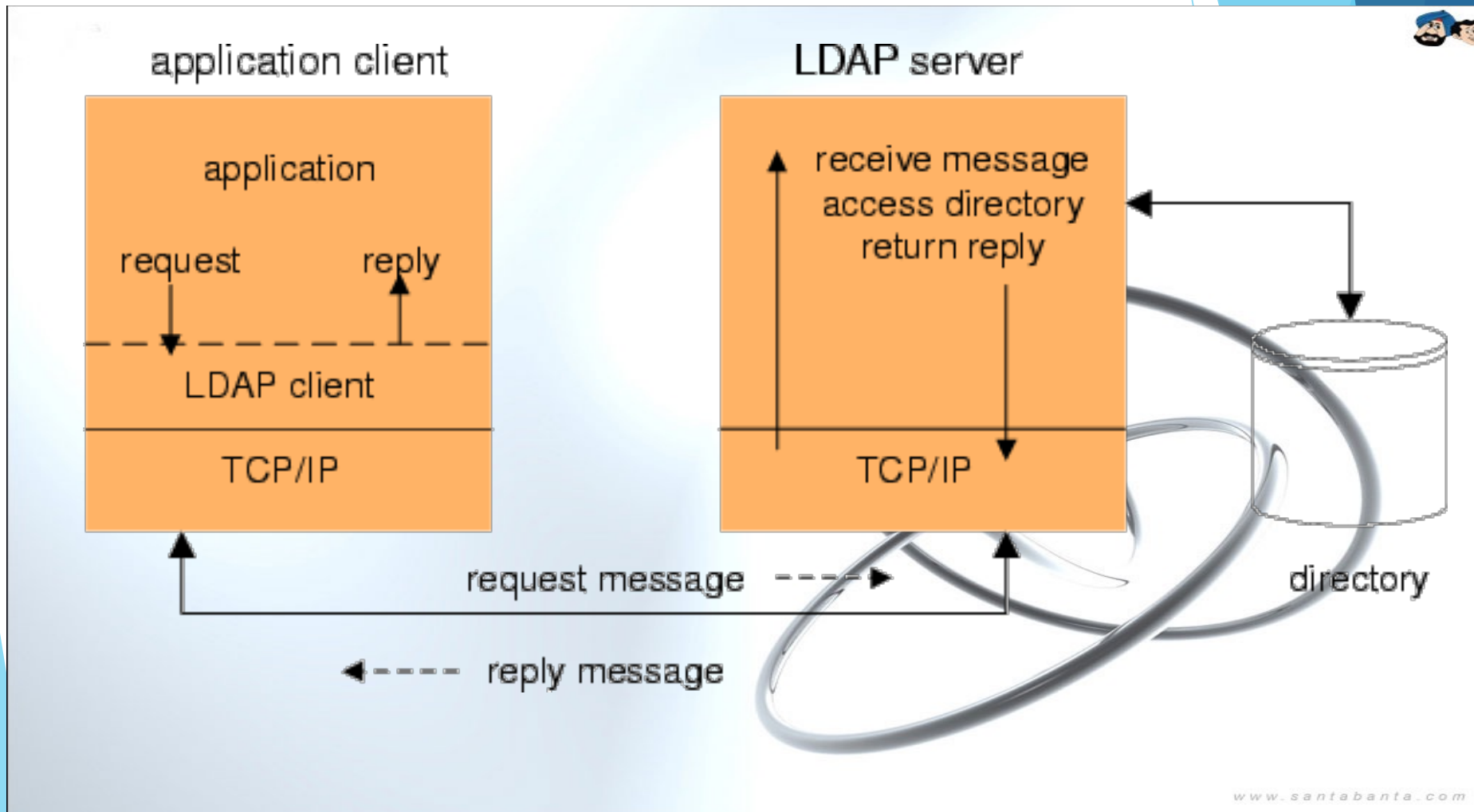
ADD/MODIFY/DELETE

- ▶ ADD request
 - ▶ Entry: LDAPDN
 - ▶ List of Attributes and values (or sets of values)
- ▶ MODIFY request
 - ▶ Used to add, delete, modify *attributes*
 - ▶ Request includes
 - ▶ Object: LDAPDN
 - ▶ List of modifications (atomic)
 - ▶ Add, Delete, Replace
- ▶ DELETE request
 - ▶ Object: LDAPDN
- ▶ MODIFY RDN: LDAPDN, newRDN, DEL_FLAG

Protocol Model

- ▶ Clients performing protocol operations against servers
 - ▶ Client sends protocol request to server
 - ▶ Server performs operation on directory
 - ▶ Server returns response (results/errors)
- ▶ Asynchronous Server Behavior

Directory Client/Server Interaction



Mapping onto Transport

- ▶ Uses Connection-oriented, reliable transport
- ▶ TCP
 - ▶ LDAPMessage PDU mapped onto TCP byte stream
 - ▶ LDAP listener on port 389
- ▶ Connection Oriented Transport Service (COTS)
 - ▶ LDAP PDU is mapped directly onto T-Data

Protocol Element Encoding

- ▶ Encoded for Exchange using BER (Basic Encoding Rules)
- ▶ BER defined in Abstract Syntax Notation One (ASN.1)
- ▶ High Overhead for BER
 - ▶ Restrictions imposed to improve perf.
 - ▶ Definite form of length encoding only
 - ▶ Bit Strings/ Octet Strings and all character string types encoded in primitive form only

LDAP Applications

- LDAP can be used for various applications, including user account management, authentication and single sign-on, address book and contact management, resource management, groupware and collaboration, and application integration.
 - LDAP enables centralized management of user accounts, authentication against the directory, and seamless access to multiple systems and applications.
 - LDAP facilitates the management of address books, contact data, and collaboration features within an organization.
 - LDAP is suitable for managing resources such as printers, storage, services, and devices, allowing centralized administration and easy access.
 - LDAP serves as a foundation for groupware and collaboration platforms, enabling team organization and information exchange.
 - LDAP can integrate different applications and systems, ensuring synchronized user data and permissions across the environment.

LDAP Implementations

- ▶ C Library API
 - ▶ LDAPv2 - RFC 1823 'The LDAP API'
 - ▶ LDAPv3 - In Internet Draft stage
- ▶ Java JNDI
- ▶ LDAP v3 uses the UTF-8 encoding of the Unicode character set.
- ▶ HTTP to LDAP gateway
- ▶ LDAP to X.500 gateway - ldapd

RFCs

- ▶ Request For Comments: RFC 4510
 - ▶ LDAP is an Internet protocol for accessing distributed directory services .
- ▶ Request For Comments: RFC 4515
 - ▶ LDAP defines a network representation of a search filter transmitted to an LDAP server.
- ▶ Request For Comments: RFC 4516
 - ▶ LDAP describes it as Uniform Resource Locator(URL).

Thank you